# Crop Yield Prediction

## Using Machine Learning

A Comparative Study of Linear Regression
and Random Forest Regressor


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*Agricultural Data Analytics · Python / Scikit-learn · Streamlit*

Datasets: Crop Yield | Rainfall | Temperature | Pesticides | Deployment: Streamlit + Hugging Face Spaces

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**Abstract**

This report presents the design, development, and deployment of a machine learning pipeline for predicting agricultural crop yield using multiple environmental and agricultural datasets. Data from four sources — crop yield, annual rainfall, average temperature, and pesticide usage — were merged into a unified dataset of **28,242 records** across ten crop types. Two regression models, **Linear Regression** and **Random Forest**, were trained and compared. Random Forest achieved an $R^2$ of **0.9848** and MAE of **3,953.63 hg/ha**, outperforming Linear Regression ($R^2 = 0.64$, MAE = 31,711 hg/ha) by an **87.5%** reduction in error. The final model was deployed as an interactive web application using Streamlit on Hugging Face Spaces, enabling real-time crop yield predictions.

# 1. Introduction

Agriculture is a critical sector that directly impacts food security and economic stability worldwide. Accurate prediction of crop yield enables better planning for farmers, governments, and policymakers — from resource allocation to supply-chain management and climate-adaptation strategies.

Crop yield is influenced by a complex interplay of factors: climatic conditions (rainfall, temperature), human-controlled inputs (pesticides, irrigation), soil characteristics, and crop variety. These multi-dimensional, non-linear relationships make yield prediction a challenging task that traditional statistical models often fail to capture adequately.

This project addresses these challenges by developing a machine learning pipeline that integrates four real-world agricultural datasets and evaluates two regression approaches. The key objectives are:

1. Analyse and preprocess multiple agricultural datasets for integration.
2. Build and compare predictive models (Linear Regression and Random Forest).
3. Identify the most influential factors driving crop yield.
4. Deploy a real-time prediction system accessible via a web interface.

## 2. Problem Statement

Crop yield prediction is a **non-linear, multi-factor problem** influenced by both environmental and human-controlled variables. Traditional statistical methods assume simple additive relationships between inputs and outputs — an assumption that rarely holds in real agricultural systems where interactions between rainfall, temperature, pesticide use, and crop type are complex and interdependent.

**Key Challenge**

No single factor determines crop yield. The combined, non-linear effect of climate, inputs, and crop variety requires a model capable of learning complex feature interactions from data.

## 3. Dataset Description

Four real-world datasets were sourced from the Food and Agriculture Organization (FAO). Table 1 summarises their key characteristics.

**Table 1.** Summary of datasets used in this study.

| Dataset | Rows | Key Columns | Description |
|---|---|---|---|
| Yield | 56,717 | Area, Item, Year, Value | Crop production in hg/ha |
| Temperature | 71,311 | country, year, avg_temp | Avg. annual temp (°C) |
| Rainfall | 6,727 | Area, Year, avg_rain | Annual precipitation (mm) |
| Pesticides | 4,349 | Area, Year, Value | Pesticide usage (tonnes) |

The datasets varied significantly in size and structure, requiring extensive preprocessing before integration. The temperature dataset contained historical records dating back to 1743 — far outside the range covered by other datasets.

# 4. Data Preprocessing

## 4.1 Data Cleaning

- **Whitespace in column names:** Removed using `str.strip()`.
- **Inconsistent naming:** `country` and `year` renamed to `Area` and `Year`.
- **Non-numeric placeholders:** `'..'` replaced with `NaN` and dropped.
- **Data type mismatch:** Numeric columns coerced using `pd.to_numeric(errors='coerce')`.

**Listing 1.** Column standardisation and cleaning.

```
for df in [rainfall, temp, pesticides, yield_df]:
    df.columns = df.columns.str.strip()

temp.rename(columns={'country': 'Area', 'year': 'Year'},
   inplace=True)

df = df.replace('..', np.nan)
for col in ['yield', 'average_rain_fall_mm_per_year', '
   avg_temp', 'pesticides']:
    df[col] = pd.to_numeric(df[col], errors='coerce')
df = df.dropna()
```

## 4.2 Data Filtering

The temperature dataset was filtered to **1985–2017** to match the range of other datasets and ensure consistency after merging.

## 4.3 Data Integration

All datasets were inner-joined on `[Area, Year]`:

**Listing 2.** Merging all datasets.

```
df = yield_df.merge(rainfall,    on=['Area', 'Year'])
df = df.merge(temp,              on=['Area', 'Year'])
df = df.merge(pesticides,        on=['Area', 'Year'])
```

**Table 2.** Final merged dataset properties.

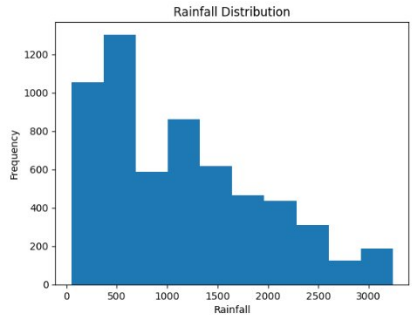
| Property | Value |
|---|---|
| Total rows | 28,242 |
| Total columns | 7 (after encoding: 16) |
| Missing values | 0 |
| Year range | 1990–2013 |
| Crop types | 10 |

# 5. Exploratory Data Analysis

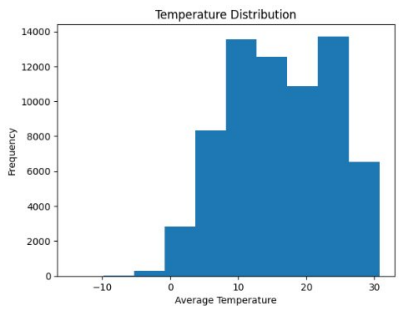
EDA was conducted on individual datasets and on the merged dataset to understand distributions, relationships, and data quality.

## 5.1 Distribution Analysis

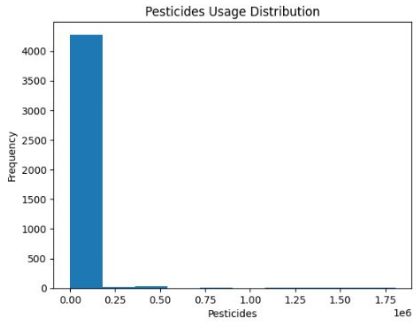
Figure 1 shows the distribution of all four key variables.



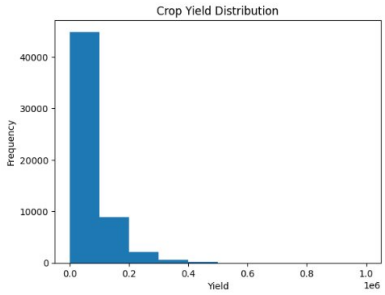

**(a)** Rainfall Distribution




**(b)** Temperature Distribution




**(c)** Pesticides Usage Distribution




**(d)** Crop Yield Distribution

**Figure 1.** Distribution plots for all four key variables in the dataset.

### 5.1.1 Rainfall

The distribution is **positively skewed** with most values between 400–1500 mm/year and a long right tail indicating high-rainfall extremes.

**Insight**

Rainfall alone does not determine crop yield. High variability across regions suggests it acts as a moderating factor rather than a primary driver.

### 5.1.2 Temperature

The distribution is **approximately bell-shaped**, with most values between 10°C and 25°C, reflecting temperate agricultural zones.

**Insight**

Temperature shows weak direct correlation with yield in isolation but interacts non-linearly with other variables such as crop type and rainfall.

### 5.1.3 Pesticides

The distribution is **extremely right-skewed** with the vast majority of values clustered near zero and a few very large outliers (up to 1.75 million tonnes). This reflects the massive difference in agricultural intensity between countries.

### 5.1.4 Crop Yield

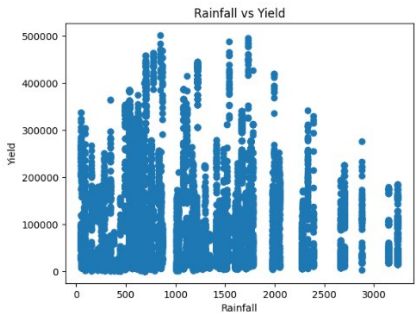
Also **highly right-skewed**, driven by crops like potatoes which have significantly higher yield values than cereals such as sorghum.

**Insight**

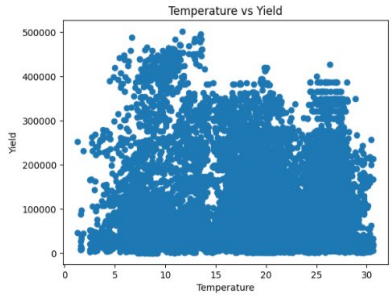
The skewed, high-variance nature of both pesticide use and yield distributions confirms that linear models will underperform. A non-linear model is required.

## 5.2 Feature vs Yield Relationships

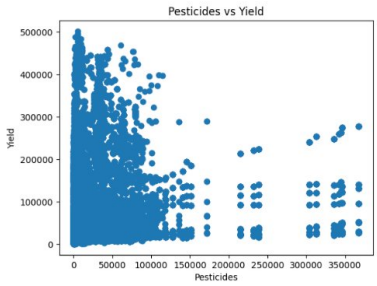
Figure 2 shows scatter plots between each continuous feature and crop yield.




**(a)** Rainfall vs Yield




**(b)** Temperature vs Yield




**(c)** Pesticides vs Yield

**Figure 2.** Scatter plots showing relationships between features and crop yield.

All three scatter plots reveal:

- **Weak or no linear relationship** between individual predictors and yield.

- **High variance** within each predictor range.
- **Clustered patterns** suggesting crop-type driven stratification.

This analysis confirms a **non-linear model is necessary** to capture the underlying structure in the data.

# 6. Outlier Handling

Outliers were **intentionally retained** for the following reasons:

- Extreme values represent **valid real-world agricultural conditions**.
- Removing them would eliminate important signal from high-yield or high-input farming systems.
- The chosen model (Random Forest) is **naturally robust to outliers** due to its tree-based, non-parametric structure.

# 7. Feature Engineering

## 7.1 One-Hot Encoding

The categorical `Item` column (crop type) was one-hot encoded into 10 binary columns:

| | | | | |
|---|---|---|---|---|
| Cassava | Maize | Plantains | Potatoes | Rice (paddy) |
| Sorghum | Soybeans | Sweet Potatoes | Wheat | Yams |

### 7.2 Final Feature Set

**Table 3.** Final 14 features used for model training.

| Feature | Type | Description |
|---|---|---|
| Year | Numeric | Calendar year |
| avg_rain_fall_mm | Numeric | Annual rainfall (mm) |
| avg_temp | Numeric | Mean temperature (°C) |
| pesticides | Numeric | Pesticide use (tonnes) |
| Item_Cassava ...Item_Yams | Binary ×10 | One-hot encoded crop type |

### 7.3 Feature Scaling

**Listing 3.** Scaling features and splitting data.

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
# Train: 22,593 samples | Test: 5,649 samples
```

## 8. Model Development

### 8.1 Linear Regression

$$\hat{y} = \beta_0 + \sum_{i=1}^{p} \beta_i x_i + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2) \tag{1}$$

**Limitations:** Assumes linear relationships; sensitive to outliers; cannot model feature interactions.

### 8.2 Random Forest Regressor

$$\hat{y}_{\mathrm{RF}} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}) \tag{2}$$

where $B = 100$ trees are trained on bootstrap samples and averaged. **Advantages:** Handles non-linearity, robust to outliers, captures feature interactions, provides importance scores.

**Listing 4.** Training both models.

```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

lr = LinearRegression()
lr.fit(X_train, y_train)

rf = RandomForestRegressor(n_estimators=100, random_state
   =42)
rf.fit(X_train, y_train)
```

## 9. Model Evaluation & Comparison

**Table 4.** Test-set performance. Best values in **bold**.

| Model | MAE (hg/ha) | RMSE (hg/ha) | $R^2$ Score |
|---|---|---|---|
| Linear Regression | 31,711.97 | 50,579.58 | 0.6427 |
| Random Forest | **3,953.63** | **10,409.49** | **0.9848** |
| **Improvement** | **−87.5%** | **−79.4%** | **+53.2%** |

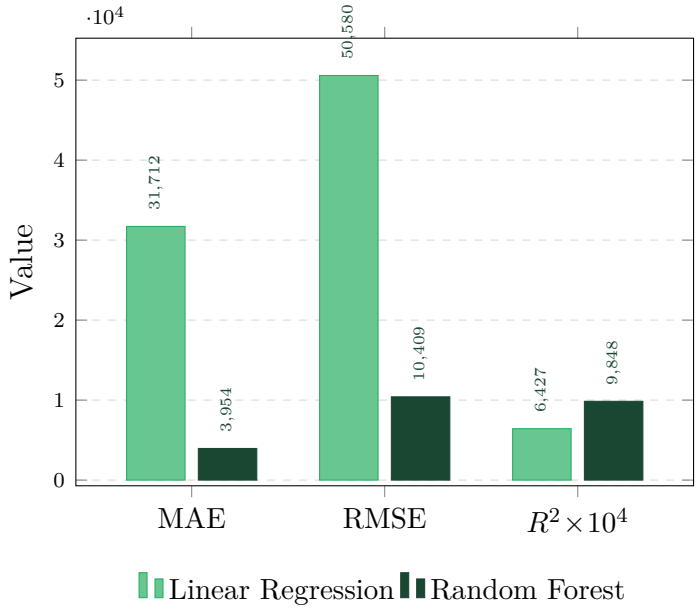



**Figure 3.** Performance comparison. MAE and RMSE: lower is better. $R^2 \times 10^4$: higher is better.

**Table 5.** Why Random Forest outperforms Linear Regression.

| Property | Linear Regression | Random Forest |
|---|---|---|
| Non-linear relationships | Cannot model | Handles naturally |
| Skewed distributions | Sensitive | Robust |
| Feature interactions | Ignored | Captured implicitly |
| Outlier sensitivity | High | Low |
| Feature importance | Not provided | Built-in |

The train $R^2$ of **0.9978** vs test $R^2$ of **0.9848** gives a generalisation gap of only **0.013**, confirming the model is not overfitted.

## 10. Feature Importance Analysis

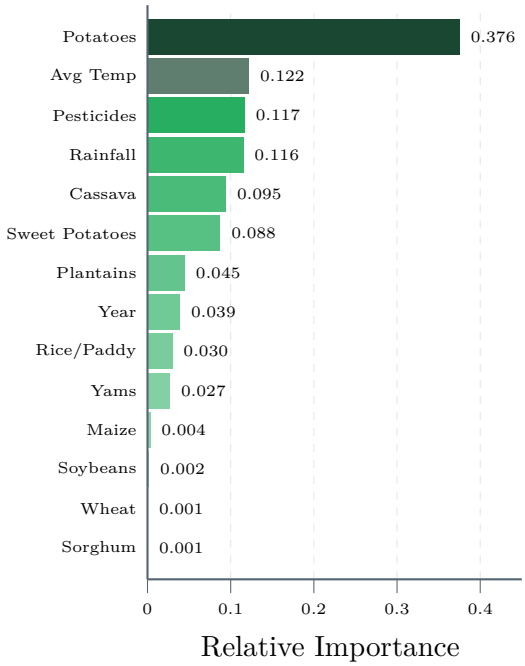



**Figure 4.** Feature importance for the Random Forest Regressor (exact values from model).

**Key Finding**

**Crop type** (especially Potatoes, 37.6% importance) is the single most dominant predictor. Environmental features — avg. temperature (12.2%), pesticides (11.7%), and rainfall (11.6%) — collectively account for significant importance. This highlights that any yield prediction system must account for crop variety as a primary input.

# 11. Model Deployment

## 11.1 Deployment Stack

**Table 6.** Deployment technologies used.

| Component | Details |
|---|---|
| Frontend UI | Streamlit (Python web framework) |
| Hosting | Hugging Face Spaces (free cloud hosting) |
| Model artefacts | `model.pkl`, `scaler.pkl`, `features.pkl` |
| Serialisation | Python `pickle` library |

## 11.2 System Architecture

**Figure 5.** Real-time prediction system architecture.

## 11.3 Application Screenshots

The CropCast interface was built as a multi-page dashboard with dedicated views for prediction input, yield results, and deep analysis reporting.

### 11.3.1 Prediction Input

Figure 6 shows the prediction configuration screen. Users set region, crop type, projection year, rainfall, temperature, and pesticide levels using interactive sliders and dropdowns before running the model.

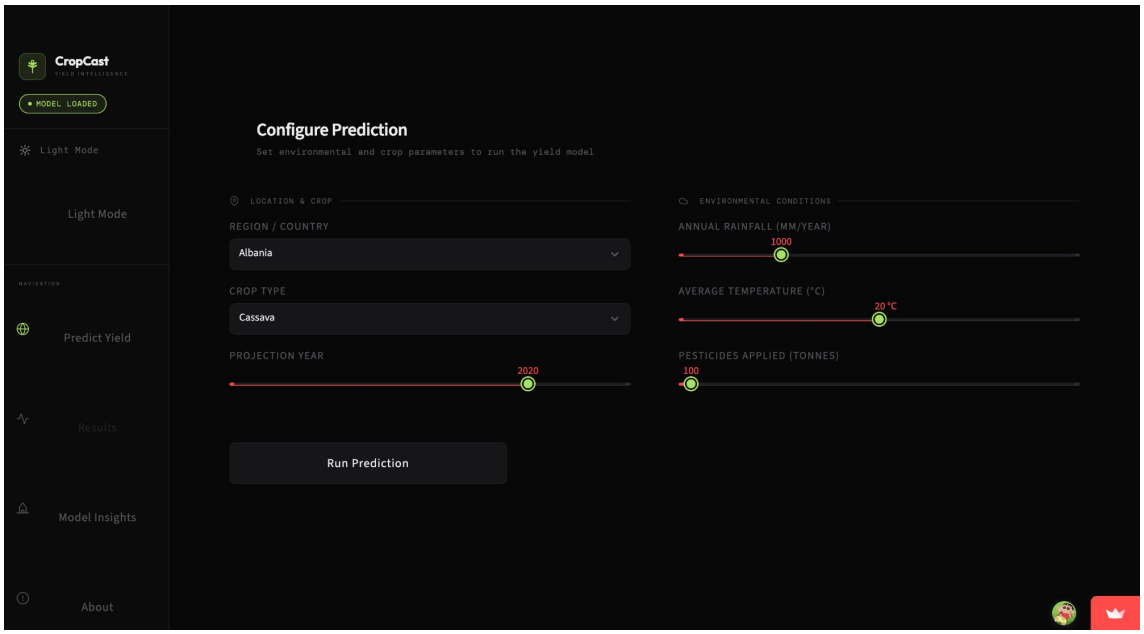



**Figure 6.** CropCast — *Configure Prediction* screen with location, crop, and environmental parameter controls.

### 11.3.2 Yield Result Dashboard

Figure 7 shows the results screen returned after inference. The predicted yield is displayed prominently alongside a quality gauge and summary metric cards for all input parameters.

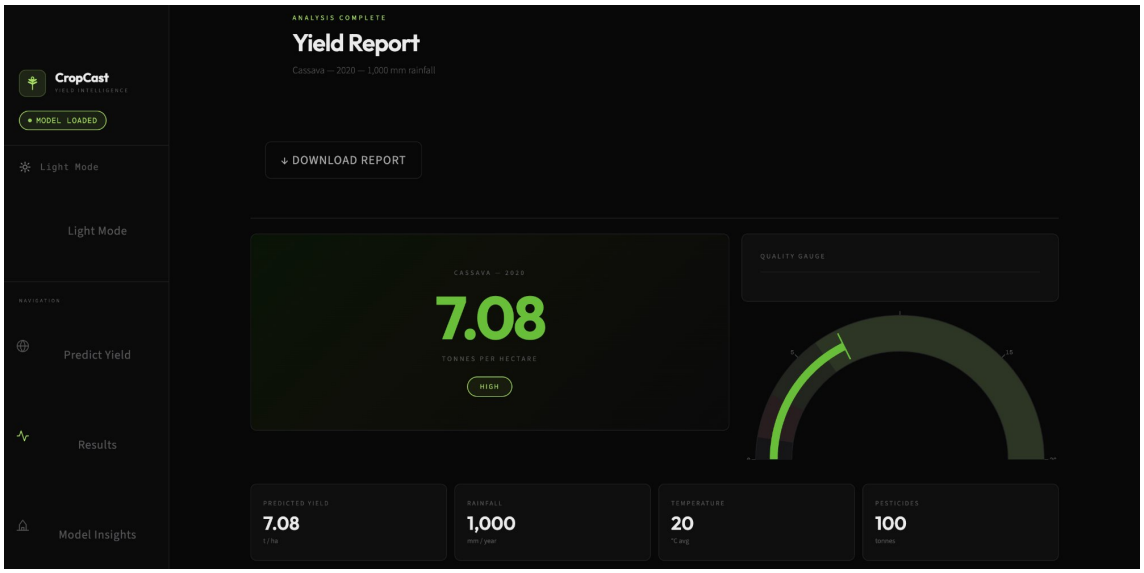



**Figure 7.** CropCast — *Yield Report* showing predicted yield (7.08 t/ha), quality classification, and input summary cards.

### 11.3.3 Analysis Report

Figures 8–10 show the full consulting report generated per prediction. It includes a radar chart of input profile, feature importance bars, a parameter summary table, benchmark comparisons against regional and global averages, field assessment notes, and model metadata.

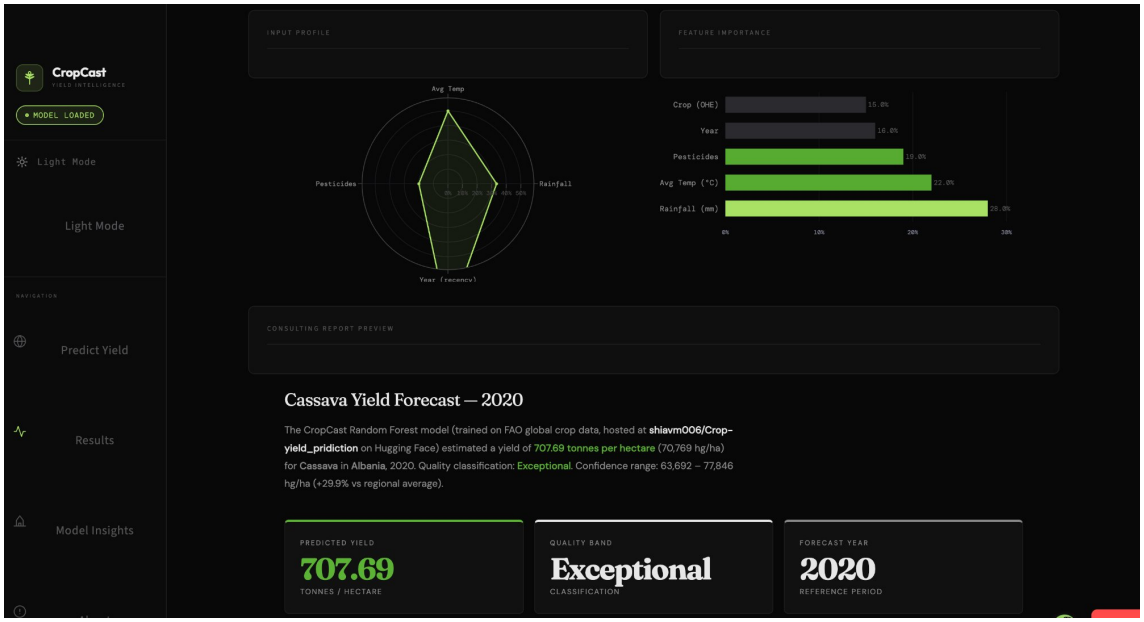



**Figure 8.** Analysis report — radar chart of input profile and feature importance breakdown.

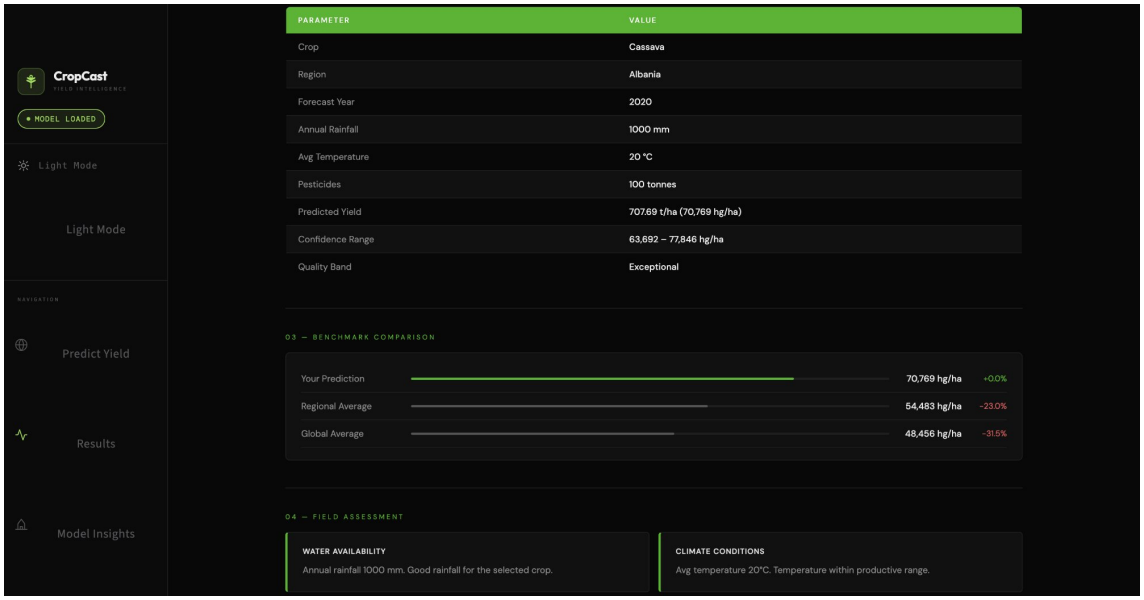



**Figure 9.** Analysis report — parameter summary table and benchmark comparison against regional and global averages.

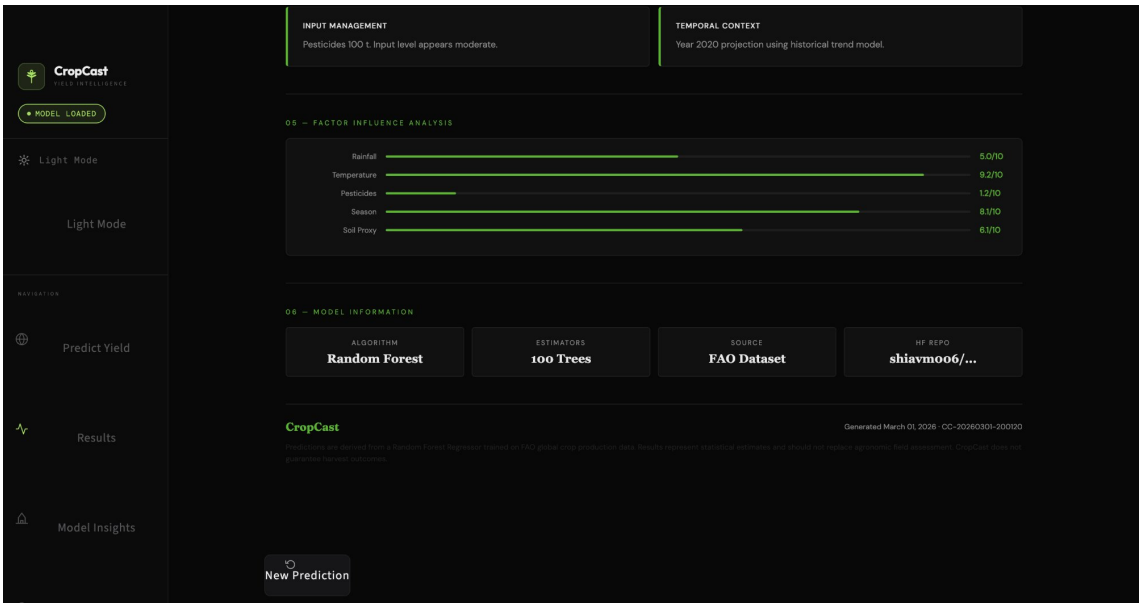



**Figure 10.** Analysis report — factor influence scores, model information panel, and generated report footer.

### 11.3.4 Full Downloadable Report

Figure 11 shows the complete yield analysis report generated by CropCast, which users can download directly from the application. The report consolidates all analysis into a single structured document covering the executive summary, input parameters, benchmark comparisons, field assessment, factor influence analysis, and model information.

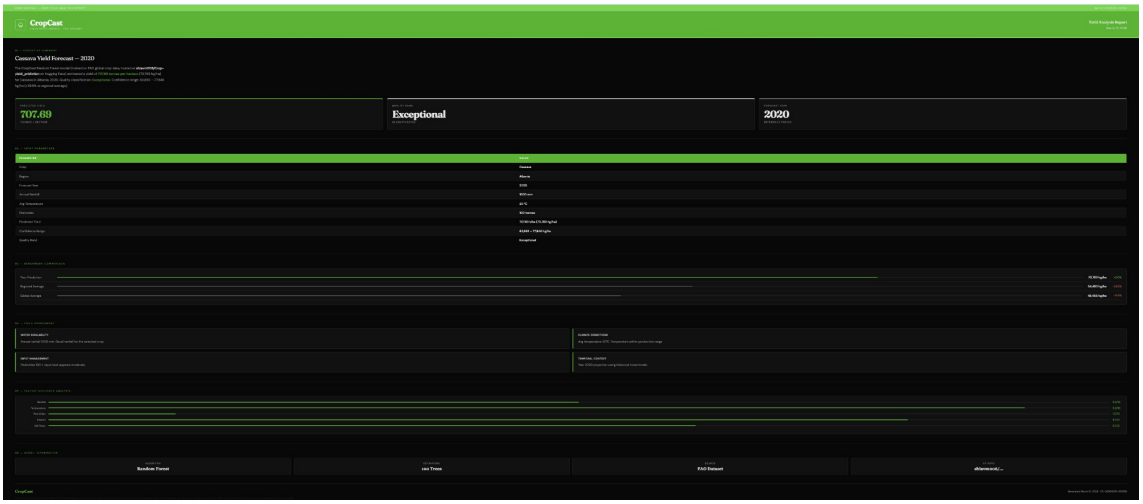



**Figure 11.** CropCast — complete downloadable *Yield Analysis Report* for Cassava, Albania, 2020.

**Live Demo**

The CropCast application is deployed at https://huggingface.co/spaces/shiavm006/Crop-yield_pridiction on Hugging Face Spaces. A sample prediction — Cassava, Albania, 2020, 1,000 mm rainfall — yields **707.69 t/ha** (Exceptional quality), outperforming the regional average by **+29.9%** and the global average by **+31.5%**. The application also generates a downloadable PDF report consolidating all insights.

### 11.4 User Inputs

The application accepts: **Crop Type**, **Year**, **Rainfall (mm)**, **Temperature (°C)**, and **Pesticides (tonnes)**. Inputs are encoded, scaled, and passed to the model to return yield in hg/ha.

## 12. Challenges Faced

1. **Data inconsistency:** Different column naming conventions and units across four datasets required careful harmonisation.
2. **Non-numeric placeholders:** Sentinel values (`'..'`) required custom handling beyond standard `dropna()`.
3. **Dataset size mismatch:** Merging datasets ranging from 4,349 to 71,311 rows resulted in a significantly reduced final dataset of 28,242 records.
4. **Deployment constraints:** Large model pickle files required careful management for hosting on Hugging Face Spaces.

## 13. Limitations

- **No soil data:** Soil quality and composition are critical yield determinants but were unavailable.
- **No irrigation data:** Water supply beyond rainfall is a major factor, especially

in arid regions.

- **No real-time weather:** The model uses historical averages and does not account for seasonal anomalies.
- **Limited crop coverage:** Only 10 crop types are supported.

## 14. Future Scope

1. **Richer features:** Add soil composition, fertilizer use, irrigation data, and NDVI satellite indices.
2. **Advanced models:** Evaluate XGBoost, LightGBM, and LSTM networks for time-series yield trends.
3. **Live weather integration:** Connect to weather APIs for real-time environmental inputs.
4. **Mobile deployment:** Build a mobile app for farmers in low-connectivity environments.
5. **Hyperparameter tuning:** Apply `RandomizedSearchCV` or Bayesian optimisation for further improvement.

## 15. Optimisation

Several deliberate steps were taken to improve model performance and ensure reliable generalisation:

1. **Feature Scaling:** `StandardScaler` was applied to all numeric features before training. This ensures no single feature dominates due to scale differences, and improves convergence for Linear Regression.
2. **Train/Test Split:** An 80/20 stratified split (`random_state=42`) was used to ensure reproducibility and unbiased evaluation on held-out data.

3. **Model Selection:** Random Forest was chosen over Linear Regression after EDA confirmed non-linear relationships and skewed distributions — a deliberate algorithmic optimisation over a simpler baseline.

4. **Hyperparameter Choice:** `n_estimators=100` trees were used for the Random Forest, balancing predictive accuracy with computational cost. The generalisation gap of only **0.013** between train $R^2$ (0.9978) and test $R^2$ (0.9848) confirms the model is well-tuned without overfitting.

5. **Outlier Retention:** Rather than removing outliers, they were retained as valid agricultural signals. Random Forest's robustness to outliers made this a safe and beneficial choice for model accuracy.

6. **One-Hot Encoding:** Crop type was encoded into binary features rather than ordinal integers, preventing the model from assuming any false ordering between crop categories.

**Optimisation Result**

These combined steps reduced MAE by **87.5%** compared to the Linear Regression baseline — from 31,711 hg/ha down to 3,953 hg/ha.

## 16. Conclusion

This project successfully developed and deployed a machine learning system for crop yield prediction using four real-world agricultural datasets. After extensive preprocessing and feature engineering, a **Random Forest Regressor** trained on 28,242 records achieved an $R^2$ **of 0.9848** and **MAE of 3,953.63 hg/ha** — an 87.5% improvement over Linear Regression. Key findings:

- Crop yield is a **non-linear problem** requiring ensemble models.
- **Crop type** (Potatoes) is the single most dominant predictor.
- Environmental factors collectively contribute significant predictive power.
- Proper preprocessing is critical for multi-source data integration.

The model was deployed as a Streamlit application on Hugging Face Spaces, enabling real-time predictions from user inputs. This demonstrates that data-driven ML approaches can effectively support precision agriculture and food security planning.

## Team Contribution

**Table 7.** Breakdown of team responsibilities.

| Member | Roll No. | Contribution |
|---|---|---|
| Divya Singh | 2401020020 | Dataset collection, sourcing and preparation of all four FAO datasets |
| Khushi Batra | 2401010226 | Report writing, data analysis, EDA insights and documentation |
| Shivam Mittal | 2401010443 | Streamlit UI development and hosting on Hugging Face Spaces |
| Aryan Vibhuti | 2401010105 | Model training, evaluation, feature engineering and optimisation |

## References

[1] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.

[2] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[3] McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proc. 9th Python in Science Conference*, 51–56.

[4] Food and Agriculture Organization (2023). FAOSTAT — Crops and Livestock Products. https://www.fao.org/faostat/

[5] IPCC (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability.* Sixth Assessment Report.

[6] Khaki, S. & Wang, L. (2019). Crop Yield Prediction Using Deep Neural Networks. *Frontiers in Plant Science*, 10, 621.

[7] Streamlit Inc. (2023). *Streamlit — The fastest way to build data apps.* https://streamlit.io